**Addis Ababa University**

College of Technology and Built Environment
School of Information Technology and Engineering
Department of Software Engineering

# SOFTWARE REQUIREMENTS SPECIFICATION

## LLM-Based Support Bot for the Ethiopian Revenue Authority

*Enhancing Information Desk Services for Taxpayers and Business Owners*

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# Revision History

| Version | Date | Author | Description |
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| 0.1 | 25 Nov 2025 | All Members | Initial draft — problem statement and scope |
| 0.5 | 10 Dec 2025 | Amanuel A. | Functional requirements and use cases draft |
| 0.8 | 18 Dec 2025 | Basliel S. | Non-functional requirements; architecture section |
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# Document Approval

| Role | Printed Name | Signature | Date |
|---|---|---|---|
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## Definitions, Acronyms, and Abbreviations

| Term / Acronym | Definition |
|---|---|
| API | Application Programming Interface — a set of protocols enabling software-to-software communication. |
| BERTScore | An automatic NLP evaluation metric that computes similarity between generated and reference text using contextual BERT embeddings. |
| BM25 | Best Match 25 — a probabilistic keyword-based ranking function used for full-text document retrieval. |
| Chunking | The process of splitting large documents into smaller, semantically coherent segments (e.g., 512–1024 tokens) for embedding and retrieval. |
| Confidence Score | A numerical value (0–1) estimating how reliably the retrieved context supports the generated answer. Scores below 0.70 trigger a fallback response. |
| Cron Job | A time-based scheduler on Unix-like systems; used here to trigger the daily knowledge-base scraping pipeline at 00:00 EAT. |
| Embedding | A fixed-size numerical vector representing the semantic meaning of a text segment; used for similarity-based retrieval. |
| EAT | East Africa Time — UTC+3, the time zone of Ethiopia. |
| ETB | Ethiopian Birr — the official currency of Ethiopia. |
| F1 Score | Harmonic mean of precision and recall; used to evaluate response quality against gold-standard answers. |
| Hallucination | A phenomenon where an LLM generates plausible-sounding but factually incorrect output not grounded in the supplied context. |
| LangChain | An open-source orchestration framework for building LLM-powered applications, used here for RAG pipeline management. |
| LLM | Large Language Model — a deep-learning model (e.g., Gemini 2.5 Flash) trained on massive corpora to understand and generate natural language. |
| MoR | Ministry of Revenue — the Ethiopian government authority responsible for tax collection and administration. |
| NLP | Natural Language Processing — a field of AI enabling computers to understand and generate human language. |
| OCR | Optical Character Recognition — conversion of scanned image-based text into machine-readable text. |
| PII | Personally Identifiable Information — data that can identify an individual (e.g., TIN, name, phone number). |
| pgvector | An open-source PostgreSQL extension that adds native vector similarity search capabilities. |
| RAG | Retrieval-Augmented Generation — an AI pattern that augments LLM generation with dynamically retrieved external knowledge. |

| | |
|---|---|
| RRF | Reciprocal Rank Fusion — a score-merging algorithm that combines ranked lists from multiple retrieval methods into a single ranked result. |
| SIGTAS | Standard Integrated Government Tax Administration System — the MoR's internal legacy tax-processing platform. |
| SRS | Software Requirements Specification — this document. |
| TIN | Taxpayer Identification Number — a unique number assigned by the MoR to identify taxpayers. |
| UI/UX | User Interface / User Experience — the visual and interaction design of the system. |
| VAT | Value Added Tax — a consumption tax levied at each stage of the supply chain, regulated by Proclamation No. 285/2002. |
| Vector DB | A database (e.g., PostgreSQL + pgvector) optimised for storing and querying high-dimensional vector embeddings. |

# Declaration

We, the undersigned members of Group 13, declare that this Software Requirements Specification represents our own original work. Where ideas or intellectual content from external sources have been incorporated, we have ensured proper citation and referencing in accordance with established academic conventions. We affirm that we have upheld all principles of academic honesty and integrity throughout the preparation of this document. We understand that any violation of these principles constitutes academic misconduct subject to disciplinary action by the institution.

| Name | Student ID | Signature | Date |
|---|---|---|---|
| Abdurahman Mohammed | ATE/8901/13 | ________________ | 22 Dec 2025 |
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# 1. Introduction

This Software Requirements Specification (SRS) defines the complete functional and non-functional requirements for the SupportBot AI system — an LLM-based information desk agent for the Ethiopian Ministry of Revenue (MoR). The document is intended to serve as the authoritative reference for all subsequent design, implementation, testing, and maintenance activities. It is structured in accordance with the IEEE 830-1998 standard for software requirements specifications.

## 1.1 Purpose

The purpose of this SRS is to:

1. Define, with precision and completeness, every requirement that the SupportBot AI system must satisfy before it can be considered fit for deployment.
2. Provide a shared understanding between the development team, the project advisor, and all identified stakeholders regarding the system's intended behaviour, constraints, and success criteria.
3. Serve as the primary input artefact for the Software Design Specification (SDS), test planning, and project management activities.
4. Establish a baseline against which change requests can be formally evaluated and traced.

This document is addressed to: software engineers (backend, frontend, AI/ML), QA engineers, and the project advisor. Secondary audiences include MoR stakeholders and future maintainers of the system.

## 1.2 Problem Statement

### 1.2.1 Context and Scale

The Ethiopian Ministry of Revenue administers tax obligations for over 4.5 million registered taxpayers, with tens of thousands of new registrations occurring monthly. Ethiopian tax legislation — including proclamations on Value Added Tax, income tax, customs duty, and excise tax — is inherently complex and is subject to continuous revision through new directives and amendments. The resulting information gap between the legal text and taxpayer understanding generates a disproportionate burden on both citizens and the MoR alike.

### 1.2.2 Identified Failure Modes of Current Channels

Two primary information channels currently exist, each with demonstrable deficiencies:

**A. Physical Information Desks**

- Taxpayers must travel to MoR branch offices and endure queues that consume over 50% of their total visit time on preliminary information gathering alone, before substantive service delivery begins.
- Desk officers are overwhelmed by query volume, limiting responses to generic guidance rather than regulation-grounded, taxpayer-specific advice.
- Survey data indicates that 60–70% of taxpayers require between 3 and 5 separate office visits to correctly complete common procedures such as VAT registration or TIN correction, generating recurring compliance costs for both parties.

- Office-hour constraints (typically 08:30–17:30 EAT, Monday–Friday) deny access to taxpayers who cannot attend during working hours.

**B. e-Services Portal (eservices.gov.et)**

- The portal hosts static, infrequently updated content. Regulations that have been superseded by newer proclamations may remain visible without a deprecation notice, creating a trust deficit.
- The portal provides no contextual or conversational interface; users must navigate large PDF archives to locate specific clauses, a task for which most individual taxpayers lack the legal literacy.
- No personalisation is offered; a small business owner seeking Category C tax guidance receives identical information to a large enterprise corporate tax officer.

### 1.2.3 Consequences of Misinformation

The consequences of acting on incorrect or outdated tax information are severe and measurable:

- Financial penalties ranging from 10% to 100% of tax liability under the Tax Administration Proclamation No. 983/2016.
- Account freezing and legal proceedings that disrupt business operations.
- Erosion of public trust in government digital services, reducing adoption of the broader e-government agenda.

## 1.3 Proposed Solution

This project delivers SupportBot AI — an AI-powered information desk agent based on Retrieval-Augmented Generation (RAG) architecture. Unlike generic chatbot deployments, SupportBot AI is purpose-built for the Ethiopian regulatory environment with the following distinguishing characteristics:

- Regulation-grounded answers derived exclusively from official MoR proclamations and directives, with mandatory clause-level citations on every factual response, eliminating hallucination risk.
- Autonomous daily synchronisation of the knowledge base with the MoR website, ensuring the bot's information reflects the current legal landscape at all times.
- Native bilingual processing in Amharic and English, with Unicode-aware tokenisation specifically tuned for the morphological complexity of the Ethiopic script.
- Taxpayer profile-based response filtering (Individual, SME/Category C, Large Enterprise) to deliver contextually relevant guidance.
- 24/7 availability via both a web interface and a Telegram bot, eliminating the physical constraints of the current desk-based model.
- A zero-footprint privacy model for guest users: no personally identifiable information is ever requested or stored.

## 1.4 Scope

### 1.4.1 In Scope

The system SHALL:

5. Automate the ingestion of tax regulations, proclamations, and directives from mor.gov.et via a scheduled web scraper.
6. Process and respond to natural language queries in Amharic and English through both a web application and a Telegram bot interface.
7. Provide citations referencing the specific regulation title, proclamation number, and article or clause underpinning every factual answer.
8. Maintain a Knowledge Base Manager through which authorised administrators can manually upload supplementary documents outside the scheduled scraping cycle.
9. Generate a Confidence Score for each response and execute a graceful fallback when confidence falls below a defined threshold.
10. Maintain anonymised query logs for monthly performance evaluation.

### 1.4.2 Out of Scope

The system SHALL NOT:

11. Perform tax calculations, compute liability amounts, or generate tax assessment notices.
12. Enable direct tax filing, payment processing, or interfacing with SIGTAS.
13. Store, process, or transmit Personally Identifiable Information (PII) such as TINs, names, or bank details.
14. Provide binding legal advice; all responses carry an explicit disclaimer.
15. Support voice interaction in the initial release (v1.0).
16. Process languages other than Amharic and English in v1.0.

## 1.5 Project Objectives

The project is governed by the following measurable objectives:

17. **OBJ-01 — Localized Legal Accuracy:** Implement a RAG pipeline that achieves a Top-5 Retrieval Accuracy of ≥85% and a Citation Accuracy of ≥95%, with all citations traceable to officially published MoR documents.
18. **OBJ-02 — Linguistic Inclusivity:** Achieve equivalent response accuracy (within ±5% deviation) for queries submitted in Amharic versus English, with ≥80% intent classification accuracy for code-switched (mixed-language) queries.
19. **OBJ-03 — Operational Efficiency:** Demonstrate through user testing that the system reduces average information-gathering time per taxpayer query from the current baseline (typically >2 hours inclusive of travel) to under 60 seconds.
20. **OBJ-04 — Information Recency:** Establish an autonomous pipeline that detects, downloads, and indexes new MoR directives within the 00:00–06:00 EAT window, maintaining a knowledge lag of no more than 24 hours from publication.
21. **OBJ-05 — Reliability & Availability:** Achieve a production uptime of ≥99.5% and a Hallucination Rate of <5% as measured by monthly manual sampling of 100 randomly selected query-response pairs.
22. **OBJ-06 — User Adoption:** Attain a Session Success Rate of ≥80% (user query resolved without escalation to a human officer) within six months of deployment.

## 1.6 Document Overview

The remainder of this SRS is organised as follows:

23. Section 2 provides a general description of the product, including its operational context, major functions, user characteristics, and general constraints.
24. Section 3 contains all specific functional and non-functional requirements, use cases, interface definitions, and design constraints.
25. Section 4 defines the formal change management process governing modifications to this SRS.
26. Section 5 lists all referenced sources.
27. Section 6 provides supplementary appendices.

# 2. General Description

This section provides the contextual and background information necessary to understand the specific requirements set out in Section 3. It does not state formal requirements but provides the conceptual framing within which those requirements must be interpreted.

## 2.1 Literature Review

### 2.1.1 Ethiopian Digital Tax and Legal Platforms

- **Ministry of Revenue (MoR) Portal (mor.gov.et):** The portal serves as the authoritative repository for Ethiopian tax legislation. A review of its usability reveals a critical navigation gap: taxpayers seeking specific regulatory clauses must manually parse large PDF documents — an activity requiring legal literacy that most individual taxpayers do not possess. This project transforms the same source documents into a semantically searchable, conversational knowledge base.
- **Ethiopian Tax Law Assistant (2025):** A benchmark RAG-based project demonstrating the feasibility of LangChain and ChromaDB for local tax queries. Research on this system established that clause-level citations substantially increase user trust in AI-generated legal guidance — a finding that directly informs the citation requirement (FR-09) in this SRS.
- **Biritu Finance Bot:** A local Ethiopian platform providing a valuable case study for financial AI adoption. Its outcomes demonstrate that Ethiopian users respond positively to conversational agents that render complex financial regulations in accessible Amharic and English dialogue, validating the multilingual design of SupportBot AI.

### 2.1.2 Amharic NLP Research

- **Linguistic Challenges (AAU NLP Research):** Research from Addis Ababa University underscores the morphological complexity of the Ethiopic script, where a single root word can yield dozens of conjugated forms. This SRS mandates Unicode-aware tokenisation and the use of multilingual pre-trained models (XLM-RoBERTa, multilingual-e5-large) as recommended by this body of literature.
- **RAG vs. Fine-Tuning for Low-Resource Languages:** Current research on low-resource language NLP consistently concludes that RAG architectures outperform fine-tuned models for legal question answering when the language has limited curated training corpora. For Amharic, fine-tuning a domain-specific legal LLM is computationally prohibitive and produces inferior retrieval fidelity relative to a well-configured RAG pipeline. This SRS adopts RAG as the definitive architectural approach.

### 2.1.3 Global Comparators

- **SARS ChatBot Lwazi (South Africa):** Demonstrates measurable reduction in physical information desk traffic following AI chatbot deployment, providing a validated precedent for SupportBot AI's anticipated operational efficiency gains.
- **KRA WhatsApp Tax Invoicing (Kenya):** Illustrates the appetite for and feasibility of mobile-first tax service delivery in the East African context, supporting the inclusion of Telegram bot integration in the scope of this project.

- **IMF — AI in Tax Administration (2024):** Provides a governance framework for responsible AI deployment in revenue authorities, informing the non-functional requirements for accuracy, explainability, and PII protection specified in Section 3.4.

## 2.2 Product Perspective

SupportBot AI is a new, standalone system designed to augment — not replace — the existing service delivery infrastructure of the Ethiopian Revenue Authority. It operates at the public-facing information layer, positioned between the taxpayer and the complex regulatory corpus, without interfacing with any of the MoR's internal transactional systems.

*[ FIGURE 1: High-Level System Architecture: SupportBot AI positioned within the MoR ecosystem — INSERT IMAGE HERE ]*

*Figure 1: High-Level System Architecture: SupportBot AI positioned within the MoR ecosystem*

The system's principal external relationships are:

28. **Data Source:** The MoR public website (mor.gov.et) serves as the exclusive authoritative source of truth. The system ingests content via an automated scraper; it does not write to or otherwise modify the MoR's infrastructure.
29. **User Channels:** Taxpayers and business owners interact with the system through a public-facing web application and a Telegram bot. Both channels expose identical functional capabilities.
30. **AI Services:** The system consumes a Large Language Model API (Gemini 2.5 Flash as primary, with a fallback to a locally hosted Llama 3 instance) and uses the multilingual-e5-large embedding model for vector generation.
31. **Administrative Interface:** Authorised MoR staff access a password-protected Admin Dashboard for knowledge base management, system monitoring, and analytics review.

## 2.3 Product Functions

At the highest level of abstraction, SupportBot AI performs six principal functions:

32. **F-1  Knowledge Acquisition:** Automated daily scraping of the MoR website to detect, download, and pre-process new tax documents, combined with an on-demand admin upload mechanism.
33. **F-2  Knowledge Indexing:** Extraction, chunking, embedding generation, and vector database upsert for all ingested documents, including OCR handling for scanned PDFs.
34. **F-3  Conversational Query Processing:** Multi-turn conversation management, language detection, intent classification, hybrid retrieval (BM25 + dense vector search with Reciprocal Rank Fusion), confidence scoring, and LLM-based response generation.
35. **F-4  Cited Response Delivery:** Display of natural language answers accompanied by mandatory source citations (document title, proclamation number, article number), rendered in the user's selected language.
36. **F-5  Session & User Management:** Guest session management (RAM-only, zero-footprint), registered user account management (persistent chat history), and taxpayer profile-based response filtering.

37. **F-6 Administration & Analytics:** Admin dashboard providing knowledge base management, system health monitoring, query log review, and monthly evaluation reporting.

## 2.4 User Characteristics

| User Class | Description | Digital Literacy | Primary Language | Core Needs |
|---|---|---|---|---|
| Guest Taxpayer (Individual) | Unregistered citizen seeking one-off information on TIN registration, income tax obligations, or basic filing procedures. | Low to Moderate | Amharic | Simple, jargon-free answers with clear procedural steps. |
| Small Business Owner (SME) | Category C taxpayer seeking guidance on VAT registration, withholding tax, trade licence renewal, and compliance deadlines. | Moderate | Amharic / Bilingual | Specific eligibility thresholds, step-by-step procedures, and deadline information. |
| Corporate Tax Officer | Large enterprise employee managing complex tax obligations including excise, customs, and transfer pricing queries. | High | English / Technical Amharic | Specific regulatory clauses, cross-proclamation references, and procedural accuracy. |
| Registered User | Any taxpayer who has created an account to access persistent chat history, saved citations, and profile-based filtering. | Moderate to High | Both | All of the above, plus history access and export capability. |
| System Administrator | MoR technical staff responsible for knowledge base management, document uploads, scraper monitoring, and system health oversight. | High | Both | Reliable dashboard, real-time system health metrics, document indexing status. |

## 2.5 General Constraints

38. **GC-01 Regulatory Adherence:** All system responses must be grounded in, and must not contradict, currently effective Ethiopian tax proclamations. The system must not generate advice that conflicts with Proclamation No. 286/2002 (Income Tax), No. 285/2002 (VAT), No. 983/2016 (Tax Administration), or their subsequent amendments.
39. **GC-02 LLM Probabilistic Nature:** Large Language Models are inherently non-deterministic. The system must compensate through mandatory confidence scoring, a fallback mechanism for low-confidence outputs, hybrid retrieval to improve document recall, and mandatory source citations to enable independent verification.

40. **GC-03 Amharic NLP Resource Scarcity:** The absence of large Amharic-specific pre-trained legal language models necessitates the use of multilingual foundation models (XLM-RoBERTa for embeddings; multilingual-e5-large). Custom model training from scratch is explicitly out of scope.
41. **GC-04 Infrastructure:** The system depends on cloud or on-premise servers capable of running Dockerized containers for the FastAPI backend, PostgreSQL database with pgvector extension, and optional GPU-accelerated inference for self-hosted LLM fallback.
42. **GC-05 Connectivity:** An active internet connection is required for full operation. Offline mode is limited to viewing previously cached conversation history.
43. **GC-06 Language Scope:** The initial release (v1.0) supports Amharic and English only. Other Ethiopian languages (Oromiffa, Tigrinya, Somali) are deferred to future releases.
44. **GC-07 No PII Collection:** The system is architecturally prohibited from requesting, storing, or transmitting any Personally Identifiable Information. This constraint shapes the design of both the guest session model and the registered user account system.

## 2.6 Assumptions and Dependencies

45. **AS-01:** The MoR website (mor.gov.et) will remain publicly accessible and will continue to publish regulatory documents in PDF or HTML format. Any structural redesign of the website that breaks the scraper's URL patterns will require a manual scraper update.
46. **AS-02:** The primary LLM (Gemini 2.5 Flash via Google AI API) will maintain its current API contract, response format, and availability SLA. A locally hosted Llama 3 instance is maintained as a fallback.
47. **AS-03:** The majority of target users possess access to a smartphone with internet connectivity and are able to use a web browser or Telegram application.
48. **AS-04:** New tax directives published on the MoR website will be in textual PDF or HTML format; fully scanned (image-only) PDFs will be the exception rather than the norm. OCR handling is included for exceptional cases.
49. **AS-05:** The PostgreSQL + pgvector extension will provide sufficient vector search performance at the target scale of ≤100 concurrent users and an expected knowledge base of up to 10,000 document chunks.

# 3. Specific Requirements

This section defines all requirements in sufficient detail to enable a software engineer to design, implement, and verify a system that satisfies them. Every requirement is assigned a unique identifier and, where applicable, a priority (High / Medium / Low) and a verification method.

## 3.1 External Interface Requirements

### 3.1.1 User Interfaces

The system shall expose the following distinct interface screens. Screen designs are presented as mockups; the actual implementation must conform to the functional requirements described in each screen's specification.

#### UI-01: Landing Page & Guest Entry

*[ FIGURE 2: Landing Page — Language Toggle, 'Start Chat as Guest' button, Login/Register panel — INSERT IMAGE HERE ]*

*Figure 2: Landing Page — Language Toggle, 'Start Chat as Guest' button, Login/Register panel*

The landing page shall display the MoR branding, a bilingual headline ('Your AI Tax Assistant / የግብር ረዳትዎ'), two call-to-action buttons ('Start Chat as Guest' and 'Log In / Register'), and a prominent disclaimer stating that the bot provides guidance only and does not constitute legal advice. A language toggle (English | አማርኛ) must be present and must persist across sessions via browser local storage.

#### UI-02: Main Chat Interface

*[ FIGURE 3: Chat Interface — Message thread, quick-action chips, Confidence Level bar, citation pill — INSERT IMAGE HERE ]*

*Figure 3: Chat Interface — Message thread, quick-action chips, Confidence Level bar, citation pill*

The chat interface shall display the conversation thread with clearly distinguished user (right-aligned) and assistant (left-aligned) message bubbles. Each assistant message must display a Confidence Level indicator (colour-coded: green ≥0.85, amber 0.70–0.84). Citations shall be rendered as clickable pill buttons displaying 'Proclamation No. XXX, Art. YY'. A persistent text input field with a file-upload icon (for document context) and a send button shall be present at the bottom of the screen. A 'Login to Save Chat' button shall be visible in the top navigation bar during guest sessions.

#### UI-03: Citation / Source Verification Panel

*[ FIGURE 4: Source Verification Side Panel — Document title, cited article, Confidence badge, Effective Date, Open PDF / Download buttons — INSERT IMAGE HERE ]*

*Figure 4: Source Verification Side Panel — Document title, cited article, Confidence badge, Effective Date, Open PDF / Download buttons*

Clicking any citation pill shall open a side panel displaying: the full document title, the specific cited article or clause (highlighted), the document effective date and status (Active/Superseded), the confidence score and its interpretation, and buttons to open the source PDF inline or download it.

#### UI-04: Settings & Language Preference

*[ FIGURE 5: Settings Screen — Language Preference (English | Amharic), Appearance (Light/Dark Mode), Text Size toggle — INSERT IMAGE HERE ]*

*Figure 5: Settings Screen — Language Preference (English | Amharic), Appearance (Light/Dark Mode), Text Size toggle*

The settings screen shall provide controls for interface language (English | Amharic), display theme (Light Mode | Dark Mode), and text size (Small | Medium | Large), with a live preview of the greeting message. A 'Save Preferences' button must commit settings to the backend for registered users, or to local storage for guest users.

#### UI-05: Admin Dashboard

*[ FIGURE 6: Admin Dashboard — KPI cards (Total Queries, Avg. Response Time, Active Sessions, Resolution Rate), Query Volume chart, System Health panel, Recent Inquiries log — INSERT IMAGE HERE ]*

*Figure 6: Admin Dashboard — KPI cards (Total Queries, Avg. Response Time, Active Sessions, Resolution Rate), Query Volume chart, System Health panel, Recent Inquiries log*

The admin dashboard shall present four KPI summary cards (Total Queries Today, Average Response Time, Active Sessions, Resolution Rate), a time-series chart of Query Volume & Response Time over the last 24 hours, a System Health panel showing real-time status of the primary server cluster, LLM API latency, and vector database availability. A Recent Inquiries table shall display the last 20 anonymised queries with columns: Timestamp, Query Snippet, Category, Status, and Sentiment score.

#### UI-06: Knowledge Base Manager

*[ FIGURE 7: Knowledge Base Manager — Document list table, drag-and-drop upload zone, Sync/Re-index button — INSERT IMAGE HERE ]*

*Figure 7: Knowledge Base Manager — Document list table, drag-and-drop upload zone, Sync/Re-index button*

The Knowledge Base Manager shall display a searchable, paginated table of all indexed documents with columns: Document Name, Type, Date Uploaded, Size, Status (Indexed | Processing | Failed), and an Actions menu (Re-index | Delete | Preview). An Upload New Regulations zone shall support drag-and-drop or file browse for PDF and DOCX files up to 25 MB. A 'Sync / Re-index All' button shall trigger a full re-indexing of the knowledge base.

### 3.1.2 Hardware Interfaces

- **HI-01 Server Side:** The application shall execute within Docker containers on virtualised cloud servers (AWS EC2 or Azure VM) or on-premise servers providing a minimum of 8 vCPUs, 32 GB RAM, and 200 GB SSD storage. GPU acceleration (NVIDIA T4 or equivalent) is required only for deployments using the self-hosted Llama 3 fallback.
- **HI-02 Client Side:** The system imposes no specific hardware requirements on client devices beyond the capability to run a modern web browser (Chrome ≥110, Firefox ≥110, Safari ≥16, Edge ≥110) or the Telegram mobile application.

### 3.1.3 Software Interfaces

| ID | Interface | Description | Protocol / Format |
| --- | --- | --- | --- |

| | | | |
|---|---|---|---|
| SI-01 | MoR Website (mor.gov.et) | Source of all tax regulatory documents. The system connects via a Python-based scraper (BeautifulSoup / Scrapy) to detect and download new PDFs and HTML pages. | HTTPS / HTML, PDF |
| SI-02 | Vector Database (PostgreSQL + pgvector) | Stores and serves vector embeddings for semantic similarity search. The pgvector extension enables cosine and dot-product similarity operations natively within PostgreSQL. | TCP / SQL |
| SI-03 | LLM Service (Gemini 2.5 Flash API) | Primary generative model for constructing natural language responses from retrieved context. Fallback: locally hosted Llama 3 via Ollama REST interface. | HTTPS / JSON (REST) |
| SI-04 | Embedding Model (multilingual-e5-large) | Converts text chunks and user queries into 1024-dimensional vectors. Served locally via a FastAPI microservice to avoid per-query API latency. | HTTP / JSON (internal) |
| SI-05 | Backend Framework (FastAPI) | Python-based REST API server handling query processing, session management, scraper orchestration, and admin operations. Orchestrated by LangChain. | HTTPS / JSON |
| SI-06 | Frontend (Next.js / React + TypeScript) | Server-side rendered web application providing the user-facing chat and admin interfaces. | HTTPS / HTML |
| SI-07 | Telegram Bot API | Telegram's Bot API enabling the @ERATaxBot Telegram interface. Communicates with the FastAPI backend via webhook over HTTPS. | HTTPS / JSON |
| SI-08 | Redis Cache | In-memory data store used for: (a) rate-limiting token buckets per IP address; (b) temporary session state for guest users; (c) caching of frequently queried embeddings. | TCP / Redis Protocol |

### 3.1.4 Communications Interfaces

- **CI-01 Transport Security:** All communications between client applications (browser, Telegram) and the server shall use HTTPS with TLS 1.2 or higher. HTTP connections shall be automatically redirected to HTTPS.
- **CI-02 API Payload Format:** All REST API requests and responses shall be formatted as UTF-8 encoded JSON. Amharic text must be represented as standard Unicode (UTF-8) without normalisation loss.
- **CI-03 Webhook (Telegram):** The system shall expose a secured webhook endpoint for inbound Telegram updates. The webhook URL shall be protected by a secret token validated against Telegram's signed request header.
- **CI-04 Internal Services:** Communication between the FastAPI backend and the embedding microservice shall occur over an internal Docker network, not exposed to the public internet.

## 3.2 Functional Requirements

Functional requirements are grouped by subsystem. Each requirement is assigned a unique ID (FR-XX), a priority level (H=High, M=Medium, L=Low), and a verification method.

### 3.2.1 Knowledge Acquisition

| ID | Requirement Statement | Priority | Verification Method |
|---|---|---|---|
| FR-01 | The system SHALL execute a scheduled scraping job daily at 00:00 EAT via a Cron scheduler. The job shall scan the MoR website's Proclamations, Directives, and Announcements pages for new or updated documents. | H | Automated test: verify cron execution log contains a success entry at 00:00 ±5 min. |
| FR-02 | The scraper SHALL compute a SHA-256 hash of each discovered document URL and file size. A document shall be downloaded only if its hash is not present in the Document Registry table, preventing redundant re-downloads. | H | Integration test: run scraper twice in succession; assert second run downloads zero new files. |
| FR-03 | The system SHALL support three document retrieval modes: (a) HTML page text extraction, (b) PDF text extraction via PyMuPDF, and (c) OCR-based extraction via Tesseract (as fallback for image-only PDFs). The extraction mode shall be selected automatically based on the PDF's text-layer presence. | H | Unit tests per extraction mode with representative fixture documents (text PDF, scanned PDF, HTML page). |
| FR-04 | The system SHALL correctly extract and preserve Amharic Unicode (Ethiopic script, Unicode block U+1200–U+137F) from all document types. Characters must survive the full pipeline from extraction through chunking to response display without corruption. | H | Test: assert round-trip Unicode integrity for a fixture document containing known Amharic strings. |
| FR-05 | Upon failure to access mor.gov.et, the scraper SHALL retry up to three times with exponential back-off (delays: 30 s, 60 s, 120 s). If all retries fail, the system SHALL log a 'Source Unreachable' entry and send an alert email to the System Administrator. No partial or corrupt data shall be committed to the database. | H | Integration test using a mock server that returns HTTP 500 for three consecutive requests; assert alert email sent and DB unchanged. |
| FR-06 | When the scraper encounters a scanned PDF for which OCR confidence is below 70%, the document SHALL be flagged as 'Requires Manual Review' in the Admin Dashboard and excluded from indexing. The admin SHALL be able to provide a corrected text file to complete the ingestion. | M | Test: ingest a deliberately low-quality scanned PDF; assert flag appears in Admin Dashboard. |
| FR-07 | The admin SHALL be able to manually trigger an ad-hoc scraping run from the Knowledge Base Manager at any time, bypassing the scheduled schedule. | M | UI test: click 'Sync / Re-index' button; verify scraper execution log shows triggered run. |

### 3.2.2 Knowledge Indexing

| ID | Requirement Statement | Priority | Verification Method |
| --- | --- | --- | --- |
| FR-08 | The system SHALL segment extracted document text into chunks of 512–1024 tokens using a sentence-boundary-aware sliding window with a 10% token overlap between consecutive chunks. The overlap preserves semantic continuity across chunk boundaries. | H | Unit test: assert no chunk exceeds 1024 tokens and each consecutive chunk pair shares ≥50 tokens. |
| FR-09 | Each chunk SHALL be stored with the following metadata: document_id, chunk_id, source_url, document_title, proclamation_number (if applicable), article_number (if applicable), effective_date, language (am\|en\|mixed), and created_at timestamp. | H | DB schema validation: assert all metadata columns are non-null for a representative set of indexed chunks. |
| FR-10 | The system SHALL generate 1024-dimensional vector embeddings for each chunk using the multilingual-e5-large model. Embeddings shall be stored in the PostgreSQL pgvector table using the cosine distance metric. | H | Integration test: assert embedding dimension is 1024 for a set of sample chunks. |
| FR-11 | The system SHALL additionally maintain a BM25 full-text search index over the chunk text, stored in PostgreSQL's built-in full-text search infrastructure. This enables hybrid retrieval. | H | Integration test: assert BM25 index returns non-empty results for a known keyword query. |
| FR-12 | When a document is re-indexed (due to admin override or content update), the system SHALL atomically delete all existing chunks and embeddings associated with that document_id before inserting the new chunks, preventing duplicate vectors. | H | Integration test: re-index a document; assert chunk count equals new chunk count (not doubled). |
| FR-13 | The system SHALL display a real-time progress indicator in the Admin Dashboard during indexing, showing current pipeline stage: Parsing \| Chunking \| Embedding \| Indexing. The document status SHALL update to 'Active/Searchable' only upon successful completion of all stages. | M | UI test: monitor Admin Dashboard during upload; assert all four stages are displayed sequentially. |

### 3.2.3 Query Processing

| ID | Requirement Statement | Priority | Verification Method |
|---|---|---|---|
| FR-14 | Upon receiving a user query, the system SHALL automatically detect the language of the input using a FastText-based language identifier. Supported languages: Amharic (am) and English (en). For code-switched inputs, the system SHALL select the dominant language for processing and log the detection as 'mixed'. | H | Unit test: assert correct language detection for a set of 20 mono-lingual and 10 mixed-language fixture queries. |
| FR-15 | The system SHALL classify the intent of each query into one of the following categories: Informational (definition or explanation of a tax concept), Procedural (step-by-step guidance for a process), Deadline (due date or calendar queries), Eligibility (threshold or qualification criteria), Penalty (consequence of non-compliance), or Out-of-Scope. Intent classification shall use the primary LLM with a structured prompt. | H | Evaluation: assert ≥90% classification accuracy on a 100-query labelled test set. |
| FR-16 | The system SHALL convert the user query into a 1024-dimensional embedding using the multilingual-e5-large model. If the user's profile is set to a specific taxpayer category, the query embedding shall be generated with a category-specific prefix prompt (e.g., 'As a Category C taxpayer: {query}') to bias retrieval toward relevant regulations. | H | Unit test: assert query embeddings are generated in under 200 ms on target hardware. |
| FR-17 | The system SHALL perform hybrid retrieval combining: (a) BM25 keyword search returning the top-20 candidate chunks, and (b) Dense vector cosine similarity search returning the top-20 candidate chunks. The two ranked lists SHALL be merged using Reciprocal Rank Fusion (RRF) with k=60 to produce a unified ranked list. | H | Integration test: assert RRF output contains no duplicates and merges contributions from both retrieval methods for queries known to benefit from each. |
| FR-18 | The system SHALL select the top-5 chunks from the RRF-merged list as the context window for LLM response generation. If fewer than 3 chunks achieve a cosine similarity score above 0.50, the system SHALL treat this as a low-confidence retrieval event and skip LLM generation, proceeding directly to the fallback response (see FR-22). | H | Integration test: submit a query entirely unrelated to tax law; assert fallback message is returned. |
| FR-19 | The system SHALL maintain multi-turn conversation context by including the last 5 message pairs (user + assistant) in the LLM prompt. Each new query SHALL be processed in the context of this conversation history to enable follow-up question handling. | H | Integration test: verify that a follow-up question ('What about businesses below that threshold?') is resolved correctly in the |

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| | | | context of the preceding answer. |

### 3.2.4 Response Generation

| ID | Requirement Statement | Priority | Verification Method |
|---|---|---|---|
| FR-20 | The system SHALL construct an LLM prompt using the following template structure: (1) System Instruction defining the bot's role and constraints, (2) Taxpayer Profile context, (3) Retrieved Context Chunks with their source metadata, (4) Conversation History, and (5) Current User Query. The prompt SHALL include an explicit instruction to the LLM not to fabricate information beyond the supplied context. | H | Manual audit: review 20 sample prompts to verify all five components are present and correctly structured. |
| FR-21 | The system SHALL compute a Confidence Score for each response as the weighted average cosine similarity of the top-5 retrieved chunks (weights: 0.40, 0.25, 0.15, 0.10, 0.10). The score shall be displayed to the user as a labelled indicator with three levels: High (≥0.85), Moderate (0.70–0.84), and Low (<0.70). | H | Unit test: assert confidence calculation matches expected values for known chunk similarity inputs. |
| FR-22 | If the Confidence Score is below 0.70 or fewer than 3 chunks meet the minimum similarity threshold (FR-18), the system SHALL NOT generate a factual response. Instead, it SHALL display the following fallback message: 'I do not have sufficient information in the current tax knowledge base to answer this question accurately. Please consult a tax officer directly at your nearest MoR branch or call [helpline number].' | H | Integration test: verify fallback triggered for a query with no relevant documents in the knowledge base. |
| FR-23 | Every factual response SHALL include at least one citation rendered as a clickable citation pill displaying: document title, proclamation number, and article number. Citations SHALL be extracted from the metadata of the retrieved chunks that contributed to the response. | H | Automated test: parse 50 sample responses and assert all contain at least one citation with the required metadata fields. |
| FR-24 | The system SHALL detect and refuse to process queries containing profane, abusive, or off-topic content (e.g., political commentary, personal insults) using a lightweight content classifier. The refusal message SHALL be polite and shall invite the user to resubmit a tax-related query. | M | Test: submit a set of flagged query fixtures; assert all are refused with the correct message. |
| FR-25 | All LLM-generated responses SHALL include the following disclaimer in the selected language: 'This information is for guidance purposes only and does not constitute legal advice. Always refer to the official MoR documents for formal proceedings.' | H | UI test: verify disclaimer is present at the bottom of every |

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| | | | assistant message bubble. |
| FR-26 | The system SHALL generate quick-action follow-up chip suggestions after each response (e.g., 'What are the penalties for late VAT filing?' / 'How do I download Form VAT-01?'). Chips shall be contextually derived from the current intent category and the cited document. | L | UI test: assert chips appear after each assistant message and submitting a chip produces a valid response. |

### 3.2.5 Session and User Management

| ID | Requirement Statement | Priority | Verification Method |
|---|---|---|---|
| FR-27 | Upon accessing the chat interface without authentication, the system SHALL immediately display a taxpayer profile selection prompt with options: 'Individual Taxpayer', 'Small Business (Category C)', 'Medium/Large Enterprise', and 'I'm not sure'. This selection shall be stored in Redis session memory for the duration of the session. | H | UI test: access chat as guest; assert profile prompt is displayed before any query can be submitted. |
| FR-28 | Guest session state (conversation history, selected profile, language preference) SHALL be stored exclusively in server-side Redis with a session TTL of 30 minutes of inactivity. No guest data SHALL be written to the persistent PostgreSQL database. All session data SHALL be irrevocably purged when the TTL expires or the browser tab is closed. | H | Security test: inspect PostgreSQL after a guest session; assert no guest messages are persisted. |
| FR-29 | The system SHALL enforce a rate limit of 15 queries per 10-minute window per unique client IP address, implemented using a Redis token-bucket algorithm. Requests exceeding this limit SHALL receive an HTTP 429 response with a Retry-After header. | H | Load test: exceed rate limit threshold; assert HTTP 429 is returned with correct headers. |
| FR-30 | A guest user SHALL be presented with a 'Login to Save This Chat' button persistently throughout their session. If the user authenticates mid-session, the system SHALL migrate the current in-memory conversation history to the user's persistent database record, maintaining continuity of the conversation. | M | Integration test: authenticate mid-session; assert full conversation history is accessible in the registered user's chat history. |
| FR-31 | Registered users SHALL be able to: (a) view all past conversation sessions with date/time labels, (b) export a conversation transcript as PDF or plain text, (c) save individual citations to a 'Saved Citations' bookmarks list, and (d) delete individual sessions or all history. | M | UI tests for each sub-requirement with a seeded test user account. |

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| FR-32 | User authentication shall be implemented using JWT (JSON Web Tokens) with a 1-hour access token TTL and a 7-day refresh token, stored in HttpOnly cookies. Passwords must be hashed using bcrypt with a minimum cost factor of 12. | H | Security test: assert JWT claims, cookie flags (HttpOnly, Secure, SameSite=Strict), and bcrypt cost factor. |

### 3.2.6 Admin and Knowledge Base Management

| ID | Requirement Statement | Priority | Verification Method |
|---|---|---|---|
| FR-33 | The Admin Dashboard SHALL be accessible only to users with the 'admin' or 'editor' role, enforced at the API gateway level. Attempting to access admin endpoints with a user-level JWT SHALL return HTTP 403. | H | Security test: assert HTTP 403 for all admin endpoints when called with a user-role JWT. |
| FR-34 | The file upload endpoint shall perform server-side validation before initiating any processing: (a) permitted MIME types: application/pdf, application/vnd.openxmlformats-officedocument.wordprocessingml.document, text/plain; (b) maximum file size: 25 MB; (c) malware scan using ClamAV. Files failing any validation SHALL be rejected with a descriptive error message; no processing pipeline stages shall be initiated. | H | Security tests: upload .exe, oversized PDF, and EICAR test virus; assert all are rejected at the validation stage. |
| FR-35 | When the system detects during upload that a document's content hash already exists in the Vector DB, it SHALL alert the admin: 'A document with identical content has already been indexed. Do you want to overwrite it?' The admin SHALL be able to confirm overwrite or cancel. An overwrite SHALL atomically delete all chunks associated with the existing document_id before re-indexing. | M | Integration test: upload the same document twice; assert duplicate detection prompt appears on second upload. |
| FR-36 | The Admin Dashboard SHALL display real-time system health metrics including: primary server CPU and RAM utilisation, LLM API response latency (rolling 5-minute average), vector DB query latency, and Telegram bot webhook status. Metric refresh interval: 30 seconds. | M | Integration test: assert metrics endpoint returns all required fields with timestamps within the last 30 seconds. |
| FR-37 | The system SHALL maintain a full audit log of all admin actions (document upload, deletion, re-indexing, manual scrape trigger) with fields: action_type, admin_user_id, target_document_id, timestamp, and result (success \| failure). Audit logs SHALL be immutable and retained for a minimum of 12 months. | H | Audit test: perform a set of admin actions; assert all are present in the audit log with correct fields. |

### 3.2.7 Logging, Analytics, and Evaluation

| ID | Requirement Statement | Priority | Verification Method |
|---|---|---|---|
| FR-38 | The system SHALL log every query-response pair with the following fields: session_id (anonymised UUID), query_text (anonymised), detected_language, classified_intent, retrieved_chunk_ids (top-5), confidence_score, response_latency_ms, user_feedback (thumbs_up \| thumbs_down \| null), and timestamp. PII scrubbing SHALL be applied before any log record is written. | H | Privacy test: assert no PII patterns (TIN format, phone numbers, email addresses) appear in any log record. |
| FR-39 | The system SHALL compute the following evaluation metrics monthly using the anonymised query log: Exact Match (EM), F1 Score, and BERTScore against a curated gold-standard test set of 100 labelled queries. Results SHALL be displayed in the Admin Dashboard's Analytics section. | M | Integration test: run evaluation pipeline on a seeded test set; assert all three metric scores are computed and stored. |
| FR-40 | The system SHALL track and report the following operational KPIs in real time on the Admin Dashboard: Total Queries Today, Average Response Time (ms), Active Concurrent Sessions, Resolution Rate (% sessions resolved without escalation), and Escalation Rate (% sessions escalated to human officer). | M | UI test: verify all five KPI cards are displayed with non-null values during active usage. |
| FR-41 | Users SHALL be able to provide feedback on each assistant response using a thumbs-up / thumbs-down control. Negative feedback SHALL optionally present a short-form categorisation (Wrong Answer \| Citation Error \| Incomplete \| Confusing Language). Feedback data SHALL be stored and surfaced in the Admin Dashboard. | L | UI test: submit negative feedback; assert feedback record appears in Admin Dashboard with correct category. |

### 3.2.8 Telegram Bot Interface

| ID | Requirement Statement | Priority | Verification Method |
|---|---|---|---|
| FR-42 | The system SHALL provide a Telegram bot (@ERATaxBot) with functional parity to the web interface: language selection, taxpayer profile selection, multi-turn conversation, cited responses, confidence indicators, and rate limiting. All processing SHALL be handled by the same FastAPI backend as the web interface, accessed via the Telegram webhook. | H | Integration test: replicate the primary use case end-to-end via the Telegram Bot API using a test bot token. |
| FR-43 | The Telegram bot SHALL support the following commands: /start (initialise session and display welcome message), /language (toggle between Amharic and English), /profile (select or change taxpayer profile), /help (display available commands), and /feedback (submit a feedback rating for the last response). | M | Bot command test: assert each command produces the correct response message. |

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| FR-44 | Guest sessions on Telegram SHALL follow the same RAM-only, zero-footprint policy as the web interface. Session state SHALL be stored in Redis keyed by Telegram chat_id, with the same 30-minute inactivity TTL. | H | Privacy test: assert no Telegram chat data is persisted to PostgreSQL for guest sessions. |

## 3.3 Use Cases

Three primary use cases are described in full. The use case diagram in Figure 8 provides a comprehensive overview of all actor-system interactions.

*[ FIGURE 8: SupportBot AI Use Case Diagram — Guest User, Registered User, Admin, and System (Cron) actors with all use cases — INSERT IMAGE HERE ]*

*Figure 8: SupportBot AI Use Case Diagram — Guest User, Registered User, Admin, and System (Cron) actors with all use cases*

### 3.3.1 Use Case UC-01: Taxpayer Enquires about Tax Regulations

| **Use Case ID** | UC-01 |
|---|---|
| **Title** | Taxpayer Enquires about Tax Regulations |
| **Actor(s)** | Primary: Guest User or Registered TaxpayerSecondary: LLM Service (SI-03), Vector Database (SI-02) |
| **Goal** | The user obtains an accurate, cited, natural-language answer to a tax-related question without visiting a physical MoR office. |
| **Priority** | High — This is the core value-delivery use case of the system. |
| **Frequency** | Continuously throughout the day; expected peak of 145 queries/hour based on current physical desk volume. |
| **Preconditions** | 1. The user has accessed the web portal via a supported browser or the Telegram bot.2. The Vector DB (SI-02) is online and populated with indexed documents.3. The LLM Service (SI-03) is operational.4. The user has selected a taxpayer profile (or selected 'I'm not sure'). |
| **Postconditions (Success)** | The user receives a natural language answer with specific citations (Proclamation number, Article number) and a Confidence Level indicator. The query-response pair is logged anonymously. |
| **Postconditions (Failure)** | A. Low Confidence: The user receives a fallback message directing them to an MoR officer.B. Service Unavailable: The user receives a retry message.C. Guest session expiry: Session data is wiped; the user is prompted to restart or log in. |
| **Main Success Scenario** | 1. The user types a query in Amharic or English (e.g., 'What is the VAT registration threshold?').2. The system detects the language (FR-14) and classifies intent as 'Eligibility' (FR-15).3. The system generates a query embedding using multilingual-e5-large (FR-16).4. The system performs BM25 + dense vector hybrid retrieval and merges results via RRF (FR-17).5. The system selects the top-5 chunks and calculates the Confidence Score as 0.91 (High) |

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| | (FR-21).6. The system constructs the LLM prompt with retrieved context, conversation history, and profile (FR-20).7. The LLM generates a response: 'Businesses with an annual taxable turnover exceeding ETB 1,000,000 are required to register for VAT.'8. The system appends the citation pill: [VAT Proclamation No. 285/2002, Art. 16].9. The system displays the response with a green Confidence indicator and the disclaimer (FR-25).10. Quick-action chips appear: 'What documents do I need to register?' / 'What is the penalty for late registration?'11. The query-response pair is logged anonymously (FR-38). |
| **Alternative Flow A1** | Low Confidence / No Relevant Information (triggered at Step 5):a. Confidence Score < 0.70 or fewer than 3 chunks meet the minimum similarity threshold.b. The system suppresses LLM generation entirely to prevent hallucination (FR-22).c. The system displays: 'I do not have sufficient information in the current tax knowledge base to answer this accurately. Please consult a tax officer directly.' |
| **Alternative Flow A2** | Mixed Language Input (at Step 2):a. The language detector classifies the query as 'mixed'.b. The system selects the dominant language for processing.c. The response is generated in the dominant language, with a note: 'Responding based on [English/Amharic] interpretation of your query.' |
| **Alternative Flow A3** | LLM or Vector DB Timeout (at Steps 4–7):a. The backend does not receive a response within 10 seconds.b. The system displays: 'The knowledge base is experiencing high traffic. Please try again in a moment.'c. The failed query is logged with status 'timeout' for monitoring. |
| **Exceptions** | Rate Limit Exceeded: >15 queries per 10 minutes from the same IP — HTTP 429 with Retry-After header (FR-29).Profanity / Off-Topic: Query is refused with a polite redirection message (FR-24). |
| **Non-Functional Considerations** | Latency: Total end-to-end response time < 5 s average, < 8 s at P95 (NFR-01).Citation Accuracy: ≥95% of citations must reference the correct article (NFR-06).Amharic Encoding: Full Ethiopic script support throughout the pipeline (FR-04). |

### 3.3.2 Use Case UC-02: Automated Knowledge Base Update

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|---|---|
| **Use Case ID** | UC-02 |
| **Title** | Automated Daily Knowledge Base Update |
| **Actor(s)** | Primary: System Scheduler (Cron)Secondary: MoR Website (SI-01), PostgreSQL + pgvector (SI-02) |
| **Goal** | Automatically detect, download, extract, chunk, and index new or updated tax documents from mor.gov.et without human intervention, ensuring the knowledge base reflects the most current legal landscape within 24 hours of publication. |
| **Priority** | High — Directly supports OBJ-04 (Information Recency). |
| **Frequency** | Daily at 00:00 EAT; processing window 00:00–06:00 EAT (NFR-02). |
| **Preconditions** | 1. The host server has active internet access to https://mor.gov.et.2. The Python scraper module is correctly configured with target URL patterns.3. The PostgreSQL + pgvector database is online and writable.4. The Redis Document Registry is populated with hashes of previously indexed documents. |

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| **Postconditions (Success)** | All newly detected documents are extracted, chunked, embedded, and indexed. The knowledge base is fully up to date. An audit log entry is written with the count of new documents processed. |
| **Postconditions (Failure)** | The Administrator receives an alert email/Telegram notification specifying the failure reason. No partial or corrupt data is committed to the vector store. The previous valid state of the knowledge base is preserved. |
| **Main Success Scenario** | 1. The system scheduler triggers the scraping pipeline at 00:00 EAT (FR-01).2. The scraper connects to mor.gov.et and traverses the Proclamations and Directives page listing.3. For each discovered document, the system computes its SHA-256 hash and checks the Document Registry (FR-02).4. For each document whose hash is absent from the registry, the system downloads the file to temporary storage.5. The system selects the appropriate extraction mode — PyMuPDF for text PDFs, Tesseract OCR for scanned PDFs, or BeautifulSoup for HTML — and extracts the text (FR-03).6. Amharic Unicode integrity is verified post-extraction (FR-04).7. The system segments the extracted text into chunks (512–1024 tokens, 10% overlap) and attaches metadata (FR-08, FR-09).8. The multilingual-e5-large model generates a 1024-dimensional embedding for each chunk (FR-10).9. Chunk vectors are upserted into the pgvector table; BM25 index is updated (FR-11).10. The Document Registry is updated with the new file hash (FR-02).11. An audit log entry is written: {action: 'auto_scrape', documents_found: N, documents_indexed: M, duration_s: T}. |
| **Alternative Flow A1** | No New Documents Found (at Step 3):a. All discovered documents are present in the Document Registry.b. The scraper logs 'No updates found — knowledge base is current.' and terminates gracefully. |
| **Alternative Flow A2** | Source Website Unreachable (at Step 2):a. The scraper receives an HTTP 4xx/5xx error or a connection timeout.b. The system retries 3 times with exponential back-off: 30 s, 60 s, 120 s (FR-05).c. If all retries fail: log 'Source Unreachable'; send alert email to the System Admin; terminate without any DB modification. |
| **Alternative Flow A3** | OCR Confidence Below Threshold (at Step 5):a. Tesseract OCR reports overall confidence < 70% for a scanned document.b. The system flags the document as 'Requires Manual Review' in the Admin Dashboard (FR-06).c. The document is excluded from the indexing pipeline. No partial or low-quality data is added to the vector store. |
| **Exceptions** | Disk Full: Pipeline terminates; critical alert sent to Admin.Embedding API Rate Limit: System pauses batch processing and resumes after the rate-limit reset window. |
| **Non-Functional Considerations** | Processing Window: Full pipeline must complete between 00:00 and 06:00 EAT (NFR-02).Data Integrity: Duplicate detection via SHA-256 hash must prevent any document from being indexed more than once (NFR — Data Integrity). |

### 3.3.3 Use Case UC-03: Admin Manual Document Upload

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|---|---|
| **Use Case ID** | UC-03 |
| **Title** | Admin Manual Document Upload (Manual Override) |
| **Actor(s)** | Primary: System Administrator / Ministry Staff EditorSecondary: PostgreSQL + pgvector (SI-02), Embedding Model (SI-04) |

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| **Goal** | Manually introduce an urgent regulation, internal circular, or corrected document into the knowledge base immediately, bypassing the 24-hour automated scraping cycle. |
| **Priority** | Medium — Critical for rapid response to emergency policy changes. |
| **Frequency** | Low (on-demand, ad-hoc); expected 1–5 times per month. |
| **Preconditions** | 1. The Admin has authenticated and holds the 'admin' or 'editor' JWT role claim (FR-33).2. The document is available in a supported format (PDF, DOCX, TXT) and is below 25 MB.3. The PostgreSQL + pgvector database is online and writable. |
| **Postconditions (Success)** | The uploaded document is parsed, chunked, embedded, and indexed. It is immediately available for retrieval in user queries (< 1 minute from upload completion, per NFR — Time to Live). Document status changes to 'Active/Searchable' in the Knowledge Base Manager. |
| **Postconditions (Failure)** | The Admin receives a specific, actionable error message. The database remains in its previous valid state. |
| **Main Success Scenario** | 1. The Admin navigates to the Knowledge Base Manager section of the Admin Dashboard.2. The Admin drags and drops a file onto the upload zone or clicks 'Browse Files' to select a file.3. The client performs an immediate format and size validation and displays the filename and size.4. The Admin clicks 'Upload & Process'.5. The server performs server-side validation: MIME type check, 25 MB size limit, and ClamAV malware scan (FR-34).6. The system checks the content hash against the Document Registry; no duplicate is found (FR-35).7. The system initiates the processing pipeline, displaying a real-time progress bar through stages: Parsing → Chunking → Embedding → Indexing (FR-13).8. Processing completes successfully. The document status changes to 'Active/Searchable'.9. The Admin Dashboard shows a preview of the extracted text (first 500 characters) for verification.10. An audit log entry is written: {action: 'manual_upload', admin_id: X, document_title: Y, chunks_created: N} (FR-37). |
| **Alternative Flow A1** | Invalid File Format (at Step 5):a. The server rejects the upload and returns: 'Invalid file type. Please upload PDF, DOCX, or TXT only.'b. No processing pipeline stages are initiated. |
| **Alternative Flow A2** | Text Extraction Failure — Encoding Error (at Step 7):a. PyMuPDF or python-docx fails to extract readable text (e.g., corrupted Amharic font embedding).b. The system halts and displays: 'Encoding Error: Could not extract text. Please convert the document to standard UTF-8 and retry.'c. Document status is set to 'Failed' in the Knowledge Base Manager. |
| **Alternative Flow A3** | Duplicate Document Detected (at Step 6):a. The content hash matches an existing document_id in the registry.b. The system presents a modal: 'This document appears to be a duplicate of [Document Title]. Do you want to overwrite it?'c. Admin selects 'Overwrite': system atomically deletes existing chunks and re-indexes the new content.d. Admin selects 'Cancel': upload is discarded; database unchanged. |
| **Exceptions** | Malware Detected (ClamAV scan positive): The file is quarantined; Admin receives: 'Security Alert: This file has been flagged by the antivirus scanner and has been quarantined. A security incident has been logged.' (FR-34)Network Interruption during upload: The system retains the partially uploaded file for 15 minutes, allowing the Admin to resume without re-selecting the file. |

| **Non-Functional Considerations** | Time to Live: The document must be searchable within 60 seconds of upload completion (NFR — TtL).Feedback: Real-time progress bar must update at least every 3 seconds during embedding generation to prevent the Admin from incorrectly concluding the system has frozen. |
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## 3.4 Non-Functional Requirements

### 3.4.1 Performance

| ID | Requirement | Target / Threshold |
|---|---|---|
| NFR-01 | Average end-to-end query response latency (query submitted to response displayed) | < 5 seconds (average); < 8 seconds (P95) |
| NFR-02 | Knowledge base update pipeline completion window | 00:00–06:00 EAT |
| NFR-03 | Concurrent user support without performance degradation | ≥ 100 concurrent users |
| NFR-04 | File upload processing — time from upload completion to searchability | < 60 seconds |
| NFR-05 | Admin Dashboard metric refresh interval | ≤ 30 seconds |

### 3.4.2 Accuracy and Hallucination Prevention

| ID | Requirement | Target / Threshold |
|---|---|---|
| NFR-06 | Top-5 Retrieval Accuracy (correct source document present in top-5 results) | ≥ 85% |
| NFR-07 | Citation Accuracy (cited Proclamation/Article matches the generated answer) | ≥ 95% |
| NFR-08 | Hallucination Rate (factually incorrect responses in monthly 100-sample audit) | < 5% |
| NFR-09 | Intent Classification Accuracy across all six intent categories | ≥ 90% |
| NFR-10 | Confidence Threshold triggering fallback response | < 0.70 |
| NFR-11 | Session Success Rate (query resolved without human escalation) | ≥ 80% |
| NFR-12 | Escalation Rate (queries requiring referral to a human officer) | < 10% |

### 3.4.3 Reliability

| ID | Requirement | Target / Threshold |
|---|---|---|
| NFR-13 | System uptime (chatbot service availability) | ≥ 99.5% monthly |

| | | |
|---|---|---|
| NFR-14 | Scheduled maintenance window maximum duration per month | ≤ 4 hours |
| NFR-15 | Automatic recovery rate from transient failures (LLM timeout, DB connection error) | ≥ 95% |
| NFR-16 | Data integrity — no duplicate vectors created for the same document | 0 duplicates per indexing cycle |

### 3.4.4 Security

| ID | Requirement | Target / Threshold |
|---|---|---|
| NFR-17 | No PII (TIN, name, phone, email) stored, processed, or transmitted in logs or DB | Zero PII in any log or DB record |
| NFR-18 | All conversation logs must be anonymised before writing (UUID session reference only) | 100% compliance |
| NFR-19 | Rate limiting — guest and unregistered users | 15 requests / 10 minutes / IP (Redis token bucket) |
| NFR-20 | Transport security for all external communications | HTTPS with TLS ≥ 1.2 |
| NFR-21 | Admin authentication — JWT token TTL | Access token: 1 hour; Refresh token: 7 days |
| NFR-22 | Password hashing algorithm and minimum cost factor | bcrypt, cost factor ≥ 12 |
| NFR-23 | Uploaded file malware scanning | ClamAV scan on every upload before any processing begins |

### 3.4.5 Multilingual Accuracy

| ID | Requirement | Target / Threshold |
|---|---|---|
| NFR-24 | Response accuracy parity between Amharic and English queries | Deviation ≤ ±5% |
| NFR-25 | Intent classification accuracy for code-switched (mixed Amharic-English) queries | ≥ 80% |
| NFR-26 | Amharic Unicode (Ethiopic script U+1200–U+137F) round-trip integrity | Zero character corruption end-to-end |

### 3.4.6 Maintainability and Evaluation

| ID | Requirement | Target / Threshold |
|---|---|---|
| NFR-27 | Code style compliance — frontend (ESLint + Prettier), backend (Black + Ruff) | Zero linting errors on CI pipeline |
| NFR-28 | Automated monthly evaluation — metrics computed from anonymised query log | EM, F1 Score, BERTScore against 100-query gold-standard test set |

| | | |
|---|---|---|
| NFR-29 | Regulation version history — ability to query the system using past effective dates | Historical versions of regulations must be retrievable by effective_date |
| NFR-30 | Docker containerisation — all components must be deployable via docker-compose | docker-compose up results in a fully functional system |

## 3.5 Inverse Requirements

The following capabilities are explicitly excluded from the system in version 1.0. Implementing any of these constitutes a scope violation requiring formal change request approval (Section 4):

50. The system SHALL NOT perform tax calculations or produce numerical tax liability assessments for any taxpayer scenario.
51. The system SHALL NOT provide an interface for direct tax filing, payment submission, or any transactional interaction with SIGTAS or any other MoR backend system.
52. The system SHALL NOT support voice input or text-to-speech output.
53. The system SHALL NOT process queries in any language other than Amharic and English.
54. The system SHALL NOT provide responses that could be construed as binding legal advice. All responses must carry the standard disclaimer (FR-25).
55. The system SHALL NOT store, process, or display any PII, even if voluntarily provided by the user in a query (such PII shall be scrubbed from logs before writing — FR-38).

## 3.6 Design Constraints

56. **DC-01 Language Model Selection:** The system must use pre-trained multilingual models for both embedding (multilingual-e5-large) and generation (Gemini 2.5 Flash). Custom model training from scratch is out of scope due to Amharic data scarcity and computational constraints.
57. **DC-02 RAG over Fine-Tuning:** The architecture must follow the RAG pattern. Fine-tuning the generative model on the tax corpus is prohibited in v1.0. Rationale: RAG provides immediate recency upon knowledge base update without the prohibitive cost of re-training. Fine-tuning may be evaluated for v2.0.
58. **DC-03 Hybrid Retrieval Mandate:** The retrieval pipeline must combine BM25 keyword search and dense vector search via Reciprocal Rank Fusion. Pure semantic search is insufficient for exact legal term lookups (e.g., specific proclamation numbers); pure BM25 is insufficient for conceptual queries.
59. **DC-04 Explainability through Citations:** Every factual response must expose the chain of retrieval — the user must always be able to trace the answer back to a specific clause in a specific published document. Black-box responses without citations are prohibited.
60. **DC-05 Data Residency:** Cloud hosting infrastructure should prioritise East African or African region data centres where available (e.g., AWS af-south-1 in Cape Town). If Ethiopian or proximate data centres are unavailable, data residency implications must be documented and disclosed to the MoR stakeholder.
61. **DC-06 Containerisation:** All system components must be containerised using Docker and orchestrated via docker-compose for local development and single-host production deployment. Kubernetes orchestration is deferred to future scaling phases.

## 3.7 Project Innovations and Contributions

This project advances beyond a naive RAG implementation through the following targeted contributions to the Ethiopian e-governance and legal AI domain:

62. **INN-01 Amharic-Optimised Legal RAG Pipeline:** Integration of Unicode-aware text extraction, morphologically-appropriate tokenisation, and multilingual embedding models tuned to handle the dense inflectional morphology of the Ethiopic script within a legal regulatory context — a combination not previously demonstrated for Ethiopian tax law.
63. **INN-02 Hybrid Retrieval with RRF for Legal Text:** Application of BM25 + dense retrieval with Reciprocal Rank Fusion specifically to the mixed-language (Amharic/English) Ethiopian tax corpus, addressing both the exact-match requirements of legal clause lookup and the semantic similarity requirements of conceptual querying.
64. **INN-03 Deterministic Evaluation Framework:** A repeatable, objective evaluation framework using Exact Match, F1 Score, and BERTScore metrics against a curated gold-standard test set, replacing subjective quality assessments. This provides a foundation for longitudinal performance tracking as the knowledge base evolves.
65. **INN-04 Zero-Footprint Guest Architecture:** A RAM-only guest session model that delivers full conversational capability without capturing any user data, directly addressing the privacy concerns and trust deficit identified among Ethiopian digital service users.
66. **INN-05 Taxpayer Profile-Aware Retrieval:** Prefix-prompt engineering that biases retrieval toward proclamations and directives relevant to the user's declared taxpayer category (Individual, SME, Large Enterprise), reducing irrelevant results and improving response precision for each user segment.

## 3.8 Logical Database Requirements

### 3.8.1 PostgreSQL Relational Schema (Core Tables)

| Table | Primary Key | Key Columns | Purpose |
|---|---|---|---|
| documents | document_id (UUID) | title, source_url, proclamation_number, effective_date, file_hash, status, created_at, updated_at | Registry of all ingested source documents. |
| chunks | chunk_id (UUID) | document_id (FK), chunk_index, content_text, content_tsvector, metadata_json, embedding (vector 1024), created_at | Stores text chunks with BM25 tsvector and pgvector embeddings for hybrid retrieval. |
| sessions | session_id (UUID) | user_id (FK, nullable for guests), platform (web\|telegram), created_at, last_active, profile_type, language_pref | Session tracking for registered users; guest sessions live in Redis only. |
| messages | message_id (UUID) | session_id (FK), role (user\|assistant), content_text, detected_language, intent_label, confidence_score, latency_ms, feedback, timestamp | Anonymised query-response log for analytics and evaluation. |
| citations | citation_id (UUID) | message_id (FK), chunk_id (FK), document_id (FK), | Maps each message to its supporting source |

| | | | |
|---|---|---|---|
| | | proclamation_number, article_number, rank_in_response | chunks for audit and citation display. |
| users | user_id (UUID) | email_hash (bcrypt), role (user\|editor\|admin), created_at, last_login | Registered user accounts. Email is stored as a one-way hash; no plaintext PII. |
| audit_log | log_id (UUID) | admin_user_id (FK), action_type, target_document_id, result, metadata_json, timestamp | Immutable admin action audit trail. |

## 3.9 Technology Stack

| Layer / Component | Technology | Version / Notes |
|---|---|---|
| Frontend | Next.js (React + TypeScript) | v14+ — server-side rendering; responsive design for mobile and desktop. |
| Backend API | Python FastAPI | v0.110+ — async REST API; LangChain for RAG orchestration. |
| LLM — Primary | Gemini 2.5 Flash (Google AI API) | Via REST API; temperature=0.0 for deterministic legal responses. |
| LLM — Fallback | Llama 3.1 8B (Ollama) | Self-hosted; activated when primary API is unavailable. |
| Embedding Model | multilingual-e5-large | 1024-dim; served locally via FastAPI microservice. |
| Language Detection | FastText (lid.176.bin) | 176-language model; sub-millisecond inference. |
| Vector Database | PostgreSQL 16 + pgvector 0.7 | Native vector similarity search; eliminates need for external vector DB service. |
| Full-Text Search | PostgreSQL tsvector / tsquery | Built-in BM25-equivalent for hybrid retrieval. |
| Relational Database | PostgreSQL 16 | Primary persistent data store for all structured data. |
| Session / Cache | Redis 7.2 | Guest session state, rate-limiting token buckets, embedding cache. |
| Web Scraping | Scrapy + BeautifulSoup4 | Scrapy for scheduled crawling; BS4 for HTML parsing. |
| PDF Extraction | PyMuPDF (fitz) + Tesseract 5 | PyMuPDF for text PDFs; Tesseract for scanned image-only PDFs. |
| DOCX Extraction | python-docx | Text extraction from Word documents. |
| RAG Orchestration | LangChain v0.2+ | Pipeline management for retrieval, prompt construction, and generation. |
| Malware Scanning | ClamAV | Open-source antivirus; scans all uploaded files before processing. |

| | | |
|---|---|---|
| Containerisation | Docker + docker-compose | All components containerised; single-command deployment. |
| Telegram Integration | python-telegram-bot v21 | Webhook-based Telegram bot interface. |
| CI/CD | GitHub Actions | Automated linting, unit tests, and integration tests on push. |

# 4. Change Management Process

This section defines the formal protocol governing modifications to the scope, requirements, architecture, or technology decisions documented in this SRS. The process is designed to ensure that all changes are transparent, justified, assessed for impact, and approved by the appropriate authority before implementation proceeds.

## 4.1 Change Request Lifecycle

All proposed changes follow a five-stage lifecycle:

67. **Stage 1 — Identification:** A potential change is identified by any team member, the project advisor, or an MoR stakeholder. Triggers include: technical limitations discovered during development, new proclamations published by the MoR, advisor feedback, or peer review findings.
68. **Stage 2 — Documentation and Submission:** The proposing team member creates a GitHub Issue labelled 'change-request'. The issue must document: (a) a clear description of the proposed change; (b) the justification and business/technical rationale; (c) the impacted SRS sections and requirement IDs; (d) estimated impact on the project timeline and team workload; and (e) any risks or trade-offs introduced.
69. **Stage 3 — Team Review and Consensus:** The change is discussed at the next scheduled weekly project meeting. A formal vote is conducted: a majority of 3 out of 5 members is required to advance the change to the advisor consultation stage. In the event of a tied vote, the Project Manager (Team Lead) casts the deciding recommendation. Minor editorial corrections to the SRS (grammar, formatting, definition updates) may be applied by the nominated Documentation Lead without a formal vote.
70. **Stage 4 — Advisor Consultation and Approval:** Changes classified as Major (those affecting project scope, core architecture, primary technology choices, or the submission deadline) must be presented to the Project Advisor (Mr. Daniel Abebe) for formal approval. The presentation must include: the Technical Feasibility Assessment and the Revised Project Timeline. Advisor approval may be given verbally (documented in meeting minutes) or in writing (email or signed approval form). Changes classified as Minor (those not affecting scope, architecture, or deadlines) may be implemented upon team majority vote alone, with the advisor notified retrospectively.
71. **Stage 5 — Implementation and Update:** Upon approval, the implementing team member: (a) increments the SRS version number in the Revision History table; (b) updates all affected sections with change markers for clarity; (c) creates or updates the corresponding backlog item/sprint task in the project management tool; and (d) notifies all team members via the shared Telegram group with a summary of the change.

## 4.2 Change Classification Criteria

| Classification | Criteria | Approval Required |
|---|---|---|
| Major | Modifies the defined system scope (adds or removes a feature from the In Scope list); changes the primary LLM, embedding model, or database technology; alters the project submission deadline; affects the core RAG architecture; adds or removes a primary user channel (web / Telegram). | Advisor + Team Majority (3/5) |

| | | |
|---|---|---|
| Minor | Adds clarification or detail to an existing requirement without changing its intent; corrects typographical or grammatical errors; updates a reference URL; adjusts a non-critical threshold (e.g., chunk overlap percentage) within a defined acceptable range; adds a Low-priority functional requirement without scope implications. | Team Majority (3/5) |

## 4.3 Roles and Responsibilities

| Role | Responsibilities |
|---|---|
| All Team Members | Identify and report potential changes. Participate in change review votes. Adhere to approved changes during implementation. |
| Project Manager (Team Lead — Abdurahman Mohammed) | Log all change requests in GitHub Issues. Facilitate weekly change review discussions. Act as the primary liaison with the Project Advisor on scope changes. Resolve tied team votes with a casting recommendation. |
| Documentation Lead (Diborah Dereje) | Update the SRS document upon approval, including Revision History increment. Maintain the change log. Ensure meeting minutes capturing change discussions are archived. |
| Project Advisor (Mr. Daniel Abebe) | Provide final authority on Major changes. Ensure the project remains within the scope of the university's academic requirements. Review and sign off on revised timelines. |

## 4.4 Documentation of Changes

72. All approved changes must be recorded in the Revision History table of this SRS document within 48 hours of approval.
73. Meeting minutes from change review discussions — including vote outcomes — must be saved in the shared project repository under /docs/meeting-minutes/.
74. All GitHub Issues related to change requests must be closed only after the corresponding SRS update has been committed and reviewed by at least one other team member.

# 5. References


75. Ethiopian Tax Law Assistant (2025). RAG implementation using LangChain and LLaMA3. [Online]. Available: https://app.readytensor.ai/publications/ethiopian-tax-law-assistant-9zFCDKHMAuRd
76. PiSpace (2025). Biritu Finance Bot: Guide to Financial Wellness in Ethiopia. [Online]. Available: https://pispace.co/ai/projects/biritu-finance-bot
77. Addis Ababa University. Machine Learning Models for Amharic Clinical Chatbot/NLP. [Online]. Available: https://etd.aau.edu.et/bitstreams/b6b1e93e-7026-4d30-8824-2f776efbf401/download
78. Lewis, P., Perez, E., Piktus, A., et al. (2020). Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. Advances in Neural Information Processing Systems 33. arXiv:2005.11401.
79. International Monetary Fund (IMF) (2024). Understanding Artificial Intelligence in Tax and Customs Administrations. IMF Fiscal Affairs Departmental Paper No. 2024/006. [Online]. Available: https://www.elibrary.imf.org/view/journals/005/2024/006/article-A001-en.xml
80. South African Revenue Service (SARS). ChatBot Lwazi Enhancements. [Online]. Available: https://www.sars.gov.za/latest-news/chatbot-lwazi-enhancements/
81. Business Daily Africa. KRA targets small traders with WhatsApp tax invoicing. [Online]. Available: https://www.businessdailyafrica.com/bd/economy/kra-targets-small-traders-with-whatsapp-tax-invoicing-4800038
82. Ethiopian Federal Democratic Republic. Value Added Tax Proclamation No. 285/2002. Addis Ababa: Negarit Gazeta, 2002.
83. Ethiopian Federal Democratic Republic. Income Tax Proclamation No. 286/2002. Addis Ababa: Negarit Gazeta, 2002.
84. Ethiopian Federal Democratic Republic. Tax Administration Proclamation No. 983/2016. Addis Ababa: Negarit Gazeta, 2016.
85. Wang, L., Yang, N., Huang, X., et al. (2024). Multilingual E5 Text Embeddings: A Technical Report. arXiv:2402.05672.
86. Robertson, S., & Zaragoza, H. (2009). The Probabilistic Relevance Framework: BM25 and Beyond. Foundations and Trends in Information Retrieval, 3(4), 333–389.
87. Cormack, G. V., Clarke, C. L., & Buettcher, S. (2009). Reciprocal Rank Fusion outperforms Condorcet and individual rank learning methods. Proceedings of SIGIR 2009, pp. 758–759.
88. IEEE Std 830-1998. IEEE Recommended Practice for Software Requirements Specifications. New York: IEEE, 1998.

# 6. Appendices

## Appendix A: Complete Requirement Traceability Matrix

The following matrix maps each functional requirement to the use case(s) it supports and the non-functional requirement(s) it interacts with. This matrix enables bidirectional traceability from stakeholder objectives to verifiable requirements.

| FR ID | Requirement Summary | Supports UC | Interacts with NFR |
|---|---|---|---|
| FR-01 | Daily cron scraping job at 00:00 EAT | UC-02 | NFR-02 |
| FR-02 | SHA-256 hash-based duplicate detection | UC-02, UC-03 | NFR-16 |
| FR-03 | Multi-mode document text extraction (PDF, HTML, OCR) | UC-02, UC-03 | NFR-08 |
| FR-04 | Amharic Unicode integrity preservation | UC-01, UC-02, UC-03 | NFR-24, NFR-26 |
| FR-05 | Scraper retry with exponential back-off and admin alert | UC-02 | NFR-15 |
| FR-06 | Low-OCR document flagging for manual review | UC-02 | NFR-08, NFR-16 |
| FR-07 | Admin-triggered ad-hoc scraping | UC-02, UC-03 | NFR-02 |
| FR-08 | Sentence-boundary chunking (512–1024 tokens, 10% overlap) | UC-01, UC-02, UC-03 | NFR-06 |
| FR-09 | Chunk metadata storage | UC-01, UC-02, UC-03 | NFR-07, NFR-29 |
| FR-10 | 1024-dim embedding generation (multilingual-e5-large) | UC-01, UC-02, UC-03 | NFR-01, NFR-06 |
| FR-11 | BM25 full-text search index | UC-01 | NFR-06 |
| FR-12 | Atomic document re-indexing (delete + insert) | UC-03 | NFR-16 |
| FR-13 | Real-time indexing progress indicator | UC-03 | NFR-04 |
| FR-14 | Automatic language detection (FastText) | UC-01 | NFR-24, NFR-25 |
| FR-15 | Intent classification (six categories) | UC-01 | NFR-09 |

| | | | |
|---|---|---|---|
| FR-16 | Profile-aware query embedding with prefix prompt | UC-01 | NFR-06 |
| FR-17 | Hybrid BM25 + dense retrieval with RRF (k=60) | UC-01 | NFR-06, NFR-01 |
| FR-18 | Top-5 chunk selection with minimum similarity gate | UC-01 | NFR-08, NFR-10 |
| FR-19 | Multi-turn conversation history (last 5 pairs) | UC-01 | NFR-01 |
| FR-20 | Structured 5-component LLM prompt construction | UC-01 | NFR-08 |
| FR-21 | Weighted Confidence Score calculation and display | UC-01 | NFR-10 |
| FR-22 | Fallback response when Confidence < 0.70 | UC-01 | NFR-08, NFR-10, NFR-12 |
| FR-23 | Mandatory citation pill on every factual response | UC-01 | NFR-07 |
| FR-24 | Profanity and off-topic content detection and refusal | UC-01 | NFR-17 |
| FR-25 | Mandatory disclaimer on all assistant messages | UC-01 | — |
| FR-26 | Contextual follow-up chip suggestions | UC-01 | — |
| FR-27 | Taxpayer profile selection prompt for guests | UC-01 | NFR-09 |
| FR-28 | RAM-only (Redis) guest session with 30-min TTL | UC-01 | NFR-17, NFR-18 |
| FR-29 | Rate limiting: 15 queries / 10 min / IP | UC-01 | NFR-19 |
| FR-30 | Mid-session login with RAM-to-DB history migration | UC-01 | NFR-18 |
| FR-31 | Registered user: history access, export, bookmarks, delete | UC-01 | NFR-17 |
| FR-32 | JWT authentication with bcrypt password hashing | UC-01, UC-03 | NFR-20, NFR-21, NFR-22 |
| FR-33 | Role-based access control for admin endpoints | UC-03 | NFR-20 |
| FR-34 | Server-side file validation: MIME, size, ClamAV scan | UC-03 | NFR-23 |
| FR-35 | Duplicate upload detection with overwrite confirmation | UC-03 | NFR-16 |

| | | | |
|---|---|---|---|
| FR-36 | Real-time system health metrics in Admin Dashboard | UC-03 | NFR-05 |
| FR-37 | Immutable admin action audit log | UC-03 | NFR-18 |
| FR-38 | Anonymised query-response logging with PII scrubbing | UC-01 | NFR-17, NFR-18, NFR-28 |
| FR-39 | Monthly EM, F1, BERTScore evaluation pipeline | — | NFR-28 |
| FR-40 | Real-time KPI dashboard (5 metrics) | — | NFR-05, NFR-11, NFR-12 |
| FR-41 | User feedback (thumbs up/down) with categorisation | UC-01 | NFR-28 |
| FR-42 | Telegram bot with full feature parity to web interface | UC-01 | NFR-01, NFR-03 |
| FR-43 | Telegram bot command set (/start, /language, etc.) | UC-01 | — |
| FR-44 | Telegram guest session Redis policy | UC-01 | NFR-17, NFR-18 |

## Appendix B: Glossary of Ethiopian Tax Proclamations Referenced

| Proclamation / Directive | Short Title | Year | Relevance to SupportBot AI |
|---|---|---|---|
| Proclamation No. 285/2002 | Value Added Tax Proclamation | 2002 | Primary source for VAT registration, thresholds, exemptions, and filing obligations. |
| Proclamation No. 286/2002 | Income Tax Proclamation | 2002 | Source for individual and business income tax schedules and obligations. |
| Proclamation No. 983/2016 | Tax Administration Proclamation | 2016 | Defines penalty structures, audit procedures, dispute resolution, and taxpayer rights. |
| Proclamation No. 1188/2020 | VAT Amendment | 2020 | Updates to VAT scope and registration procedures. |
| Regulation No. 98/2019 | Excise Tax Regulation | 2019 | Applicable to manufacturers and importers of excisable goods. |

## Appendix C: API Endpoint Summary

The following table summarises the primary REST API endpoints exposed by the FastAPI backend. Full API documentation is maintained as an OpenAPI 3.0 specification at /docs in the running application.

| Method | Endpoint | Auth Required | Description |
|---|---|---|---|

| | | | |
|---|---|---|---|
| POST | /api/v1/chat/query | No (rate-limited) | Submit a user query; returns LLM-generated response with citations and confidence score. |
| GET | /api/v1/chat/session/{id} | Session Token | Retrieve conversation history for a session. |
| POST | /api/v1/auth/register | No | Register a new user account. |
| POST | /api/v1/auth/login | No | Authenticate and receive JWT access + refresh tokens. |
| POST | /api/v1/auth/refresh | Refresh Token | Obtain a new access token using a valid refresh token. |
| GET | /api/v1/admin/documents | Admin JWT | List all indexed documents with status. |
| POST | /api/v1/admin/documents | Admin JWT | Upload and process a new document. |
| DELETE | /api/v1/admin/documents/{id} | Admin JWT | Delete a document and all its associated chunks and embeddings. |
| POST | /api/v1/admin/scrape | Admin JWT | Trigger an ad-hoc scraping run. |
| GET | /api/v1/admin/health | Admin JWT | Return real-time system health metrics. |
| GET | /api/v1/admin/analytics | Admin JWT | Return aggregated query analytics and KPIs. |